HEART DISEASE RISK PREDICTION AND EXPLAINABLE HEALTHCARE ANALYTICS

A Machine Learning Case Study Using Python, Predictive Modeling, and Explainable AI

Project Domain: Healthcare Analytics and Data Science

Project Type: Supervised Machine Learning — Binary Classification

Programming Language: Python

Tools and Technologies: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, SHAP, Microsoft Power BI

Dataset: heart.csv

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1. ABSTRACT

Heart disease is a major public health concern, making cardiovascular data analysis an important area of healthcare research. Machine learning techniques provide methods for identifying patterns in historical data and evaluating whether these patterns can distinguish between different outcome categories.

This project presents a machine learning and explainable analytics framework using the `heart.csv` dataset. The primary objective is to develop and compare classification models, examine the factors influencing model predictions, and communicate the analytical findings through visualizations.

The dataset contains 303 observations, 13 input features, and one binary target variable. The features include demographic information and clinical measurements such as age, resting blood pressure, cholesterol, maximum heart rate, exercise-induced angina, and chest pain category.

The proposed methodology includes exploratory data analysis, data preprocessing, model development, cross-validation, performance evaluation, feature importance analysis, and classification threshold investigation.

Logistic Regression and Random Forest are used as the initial baseline models. Additional algorithms, including Decision Tree, Support Vector Machine, and XGBoost, can be evaluated to extend the comparison.

The models are assessed using accuracy, precision, recall, F1-score, and the Area Under the Receiver Operating Characteristic Curve. Explainability techniques, including permutation feature importance and SHAP, are proposed to investigate how trained models use the available features.

The project demonstrates an end-to-end data science workflow while emphasizing model validation, interpretability, reproducibility, and responsible use of healthcare-related data.

Keywords: Heart Disease Prediction, Machine Learning, Predictive Analytics, Explainable AI, Healthcare Analytics, Python, Classification, Data Visualization.

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2. INTRODUCTION

2.1 Background

Healthcare data contains valuable information about demographic characteristics, clinical measurements, medical histories, and health outcomes.

Analyzing this information can help researchers investigate patterns and relationships within patient populations.

Machine learning is a branch of artificial intelligence that enables computational models to learn patterns from historical observations and apply those patterns to previously unseen data.

In healthcare research, classification algorithms can be used to study whether selected characteristics help distinguish between different outcome categories.

However, building a useful predictive model requires more than training an algorithm and reporting its accuracy. A comprehensive workflow must include data quality assessment, exploratory analysis, appropriate preprocessing, model validation, performance evaluation, and interpretation.

This project applies these principles to a heart disease dataset using Python and machine learning.

2.2 Project Overview

The project investigates the relationship between the input features in `heart.csv` and its binary target variable.

The initial analysis uses Logistic Regression and Random Forest as baseline classification models.

The proposed extended analysis introduces additional machine learning algorithms and explainability methods to examine predictive performance from multiple perspectives.

The project also investigates how classification thresholds affect precision, recall, and prediction outcomes.

Finally, the findings are organized into a structured case study that can be supported by visualizations and an optional Power BI dashboard.

The project is designed for educational and portfolio purposes. It is not intended to provide medical diagnoses or replace professional healthcare assessments.

2.3 Motivation

A basic machine learning project often focuses primarily on model accuracy.

However, accuracy alone does not explain how a model behaves, which features influence its predictions, or how frequently it makes different types of errors.

A more comprehensive approach should answer the following questions:

- How effectively can different algorithms distinguish between the target classes?
- Which evaluation metrics provide the most useful information about model performance?
- How stable are the results across validation folds?
- Which features contribute most to the fitted models' predictions?
- How do different classification thresholds affect precision and recall?
- What limitations must be considered before interpreting the results?

These questions form the foundation of this case study.

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3. PROBLEM STATEMENT

The objective of this project is to investigate whether the input features in the `heart.csv` dataset can be used to classify its binary target variable.

The project also aims to compare different machine learning algorithms, evaluate their performance using multiple metrics, and investigate the features that influence their predictions.

The central research question is:

How can machine learning and explainable analytics be used to analyze a heart disease dataset, compare classification models, and interpret their predictions responsibly?

The project addresses several specific questions.

- What patterns can be identified through exploratory data analysis?
- Which classification algorithms perform best under a consistent evaluation procedure?
- How effectively do the models identify positive target cases?
- Which features contribute most to the models' predictions?
- How do different probability thresholds affect classification performance?
- What limitations affect the reliability and generalizability of the results?

The project focuses on analytical model development and evaluation rather than establishing medical causation or developing a clinically validated diagnostic system.

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4. PROJECT OBJECTIVES

The primary objectives of the project are:

- Analyze the structure and characteristics of the heart disease dataset.
- Perform exploratory data analysis to understand feature distributions and relationships.
- Investigate missing values, duplicate records, and potential data quality issues.
- Prepare the dataset for supervised machine learning.
- Develop a baseline classification model using Logistic Regression.
- Compare the baseline with Random Forest and additional classification algorithms.
- Evaluate performance using accuracy, precision, recall, F1-score, and ROC-AUC.
- Apply stratified cross-validation to investigate model stability.
- Analyze feature importance using appropriate interpretation techniques.
- Investigate the effects of different classification thresholds.
- Create visualizations to communicate the analytical findings.
- Design an optional dashboard for exploring the dataset and model results.
- Document the limitations and responsible use of the models.

5. PROJECT SCOPE

The project combines data analysis, machine learning, model evaluation, and explainable analytics.

5.1 Data Analysis

The data analysis component includes:

- Dataset inspection and summary statistics.
- Data quality assessment.
- Target distribution analysis.
- Feature distribution analysis.
- Correlation analysis.
- Investigation of relationships between selected features and the target.

5.2 Machine Learning

The machine learning component includes:

- Logistic Regression.
- Decision Tree.
- Random Forest.
- Support Vector Machine.
- Optional XGBoost implementation.
- Data preprocessing and model training.
- Cross-validation.
- Hyperparameter tuning.
- Final model evaluation.

5.3 Explainable Analytics

The explainability component investigates how trained models use the input features.

It may include:

- Model-specific feature importance.
- Permutation feature importance.
- SHAP-based interpretation.
- Analysis of individual predictions.

5.4 Visualization and Reporting

The reporting component includes exploratory charts, model performance visualizations, confusion matrices, ROC curves, and feature importance plots.

The project can also be extended with a Power BI dashboard to communicate the results interactively.

The scope is limited to the supplied dataset and the experiments performed on it. Clinical validation, medical deployment, and real-world diagnostic use are outside the current scope.

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6. DATASET DESCRIPTION

6.1 Dataset Overview

The project uses a CSV file named `heart.csv`.

The supplied dataset contains 303 observations and 14 columns, consisting of 13 input features and one target variable.

Each row represents an observation, while the columns describe demographic characteristics and clinical measurements.

The target variable is named `target`.

6.2 Dataset Characteristics

The initial dataset inspection identifies the following characteristics:

- Total observations: 303.
- Number of input features: 13.
- Number of target variables: 1.
- Total columns: 14.
- Observations with target value 0: 138.
- Observations with target value 1: 165.
- Missing values in the supplied CSV: None detected.

Approximately 45.5% of observations belong to target class 0, while approximately 54.5% belong to target class 1.

This indicates that the two classes are relatively balanced in the supplied dataset.

Nevertheless, the dataset is small, so its performance estimates may vary depending on how the data is divided for training and testing.

6.3 Feature Description

The dataset contains the following input features.

Age: Represents the individual's age.

Sex: A numerically encoded sex category.

Chest Pain Category: Represents the chest pain classification recorded in the dataset.

Resting Blood Pressure: Represents resting blood pressure measurements.

Serum Cholesterol: Represents the recorded serum cholesterol measurement.

Fasting Blood Sugar: Indicates a fasting blood sugar category based on the dataset's encoding.

Resting ECG: Represents the recorded resting electrocardiographic category.

Maximum Heart Rate: Represents the maximum heart rate achieved during the recorded assessment.

Exercise-Induced Angina: Indicates the recorded exercise-induced angina category.

ST Depression: Represents the ST depression measurement associated with exercise.

ST Segment Slope: Represents the recorded slope category.

Major Vessels: Represents the number of major vessels encoded in the dataset.

Thalassemia-Related Category: Represents the encoded category recorded under the `thal` feature.

Target: The binary outcome variable used for classification.

The exact meanings and encodings of the categorical variables should be verified against the documentation for the specific dataset version.

6.4 Target Variable

The target variable represents the outcome the classification models attempt to predict.

In commonly distributed versions of this dataset, the target distinguishes between observations associated with the absence and presence of heart disease.

The target coding must be verified against the dataset documentation before the results are interpreted clinically.

The models learn to predict the labels provided in the dataset. They do not independently establish a medical diagnosis.

6.5 Dataset Limitations

The dataset contains relatively few observations and may not represent the diversity of contemporary patient populations.

Its original collection process, population characteristics, and target definition must be considered when interpreting the findings.

Consequently, the dataset is appropriate for demonstrating machine learning methods but is insufficient on its own to establish clinical reliability.

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7. TECHNOLOGY STACK

7.1 Python

Python serves as the primary programming language for data preparation, exploratory analysis, model development, and evaluation.

Its machine learning and data analysis libraries make it suitable for developing a reproducible analytical workflow.

7.2 Pandas

Pandas is used to load the CSV file, inspect the dataset, manipulate tabular data, and calculate descriptive statistics.

7.3 NumPy

NumPy supports numerical calculations and array operations used throughout the analysis.

7.4 Matplotlib

Matplotlib is used to create visualizations such as histograms, bar charts, ROC curves, and model comparison charts.

7.5 Seaborn

Seaborn supports statistical visualizations, including correlation heatmaps and grouped distribution plots.

7.6 Scikit-learn

Scikit-learn provides tools for preprocessing, model training, cross-validation, hyperparameter tuning, and performance evaluation.

It also provides implementations of the initial classification algorithms.

7.7 SHAP

SHAP can be used to investigate how individual feature values contribute to a model's output.

It is an optional extension and should be included in the final results only after implementation and verification.

7.8 Microsoft Power BI

Power BI can be used to create an interactive dashboard that presents dataset characteristics, exploratory findings, and model performance.

The dashboard complements the Python analysis by providing a clear visual summary of the project.

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8. PROJECT METHODOLOGY

The project follows an end-to-end machine learning workflow.

Step 1: Data Collection

Load the supplied `heart.csv` file into Python.

Verify that the file contains the expected columns and target variable.

Step 2: Data Understanding

Inspect the dataset dimensions, column names, data types, and descriptive statistics.

Confirm the target encoding using the available dataset documentation.

Step 3: Data Quality Assessment

Check for missing values, duplicate observations, invalid entries, and inconsistent encodings.

Document any cleaning decisions.

Step 4: Exploratory Data Analysis

Investigate target distribution, numerical feature distributions, categorical feature relationships, and correlations.

Use visualizations to identify patterns that warrant further investigation.

Step 5: Data Preparation

Separate the input features from the target variable.

Create a stratified training and testing split.

Apply the appropriate preprocessing steps within machine learning pipelines.

Step 6: Baseline Model Development

Train Logistic Regression and Random Forest as the initial classification models.

Record the model configurations and validation procedures.

Step 7: Extended Model Development

Implement additional algorithms, such as Decision Tree and Support Vector Machine.

XGBoost may also be included if the additional implementation is completed.

Step 8: Cross-Validation and Tuning

Use stratified cross-validation on the training data to compare candidate models and tune their hyperparameters.

Avoid using the final test set for model selection.

Step 9: Model Evaluation

Evaluate the selected model on the held-out test set using multiple classification metrics.

Step 10: Explainability

Investigate model behavior through feature importance and, if implemented, SHAP explanations.

Step 11: Threshold Analysis

Evaluate how different classification thresholds affect prediction outcomes using validation data.

Step 12: Reporting

Present the verified results through visualizations, written analysis, and an optional Power BI dashboard.

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9. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis, commonly known as EDA, is used to understand the dataset before training machine learning models.

It helps identify patterns, examine feature distributions, and investigate possible data quality issues.

9.1 Target Distribution Analysis

The dataset contains 138 observations with target value 0 and 165 observations with target value 1.

A bar chart can visualize the number of observations in each class.

This helps determine whether the target variable has substantial class imbalance.

Although the classes are relatively balanced, the evaluation should still include metrics that measure performance across both classes.

9.2 Age Distribution

A histogram can be used to examine the distribution of age.

A box plot can help visualize the median, spread, and potential unusual observations.

Age distributions can also be compared across the target classes to investigate whether the supplied data contains differences between the groups.

Any observed pattern should be described as a dataset-level association rather than a causal relationship.

9.3 Resting Blood Pressure and Cholesterol

Histograms and box plots can be used to examine resting blood pressure and cholesterol measurements.

These visualizations help identify the range of observed values and potential outliers.

Unusual values should be investigated before any records are removed.

A value that appears unusual is not necessarily incorrect.

9.4 Maximum Heart Rate

The maximum heart rate feature can be examined using a histogram and grouped distribution plot.

Comparing its distribution across target classes can reveal whether the supplied observations show differences between the groups.

Such differences do not establish that maximum heart rate independently determines disease status.

9.5 Categorical Feature Analysis

Categorical or integer-encoded variables can be examined using grouped bar charts.

The analysis can investigate:

- Chest pain category.
- Resting ECG category.
- Exercise-induced angina.
- ST segment slope.
- Thalassemia-related category.

The proportion of each target class within these categories can be compared.

The interpretations should account for the meaning of the encoded variables and the limitations of the dataset.

9.6 Correlation Analysis

A correlation heatmap can be used to examine relationships between numerical variables.

This visualization can help identify highly correlated features and potential redundancy.

However, integer-encoded categories may not have meaningful numerical distances. Their correlations should therefore be interpreted cautiously.

Correlation does not establish causation.

9.7 Missing Values and Duplicate Records

The initial inspection found no missing values in the supplied CSV.

Duplicate records should still be investigated.

If duplicates are found, their origin should be examined before removal because repeated observations may have different explanations.

The final report should document the actual findings from the completed data quality checks.

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10. DATA PREPROCESSING

Data preprocessing prepares the dataset for machine learning and helps reduce avoidable errors during model development.

10.1 Separating Features and Target

The dataset is divided into two components.

- **Input features:** The 13 predictor variables.
- **Target variable:** The `target` column.

The target column must not be included among the predictors because doing so would introduce target leakage.

10.2 Train-Test Split

An 80:20 stratified train-test split can be used to separate model development from final evaluation.

With 303 observations, this produces approximately 242 training observations and 61 testing observations.

Stratification helps preserve the approximate proportions of the target classes.

A fixed random state can be used to improve reproducibility.

10.3 Feature Scaling

Feature scaling is important for algorithms such as Logistic Regression and Support Vector Machine.

StandardScaler can be used to standardize the numerical input features within a pipeline.

Tree-based models, such as Decision Tree and Random Forest, generally do not require feature scaling.

10.4 Categorical Encoding

Some variables are categorical even though they are represented by integers.

The preprocessing approach should reflect the meaning of each feature.

Where appropriate, one-hot encoding may be considered for categorical predictors.

The selected encoding should be documented and applied consistently.

10.5 Data Leakage Prevention

Data leakage occurs when information unavailable at prediction time enters model training or evaluation.

To reduce this risk:

- Split the data before fitting preprocessing transformations.
- Fit scaling and encoding operations using training data only.
- Use pipelines where appropriate.
- Perform hyperparameter tuning within cross-validation.
- Keep the final test set separate until the evaluation stage.

These practices help produce more reliable estimates of model performance.

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11. MACHINE LEARNING MODELS

The project begins with two baseline algorithms and proposes additional models for comparison.

11.1 Logistic Regression

Logistic Regression is a supervised classification algorithm that estimates the probability of belonging to a particular class.

It provides a useful baseline for binary classification problems.

Advantages

- Relatively simple to implement.
- Supports probability estimation.
- Can be easier to interpret than more complex models.
- Provides a useful reference for comparing other algorithms.

Limitations

- May not capture complex nonlinear relationships without additional feature engineering.
- Can be affected by multicollinearity.
- Its performance depends on the relationship between the predictors and the target.

11.2 Decision Tree

Decision Tree is a classification algorithm that divides observations through a sequence of feature-based decisions.

Advantages

- Can model nonlinear relationships.
- Produces understandable decision rules.
- Generally does not require feature scaling.

Limitations

- Can overfit the training data.
- May be sensitive to small changes in the dataset.
- A single tree may generalize less effectively than an ensemble model.

11.3 Random Forest

Random Forest combines predictions from multiple decision trees.

It can capture nonlinear relationships and interactions between input features.

Advantages

- Can model complex patterns.
- Provides a useful comparison with linear models.
- Generally does not require feature scaling.
- Supports model-specific feature importance analysis.

Limitations

- Is more difficult to interpret than an individual decision tree.
- Can overfit under some configurations.
- Feature importance may be affected by correlations between predictors.

11.4 Support Vector Machine

Support Vector Machine attempts to separate classes using a decision boundary.

Different kernel functions allow it to model linear or nonlinear boundaries.

Advantages

- Can perform well on small and medium-sized datasets.
- Supports different kernel functions.
- Can model nonlinear decision boundaries.

Limitations

- Requires appropriate feature scaling.
- Performance depends on the choice of kernel and hyperparameters.
- Probability estimation may require additional calibration.

11.5 XGBoost

XGBoost is a gradient-boosting algorithm that combines multiple decision trees.

It builds trees sequentially to improve predictive performance.

Advantages

- Can capture complex patterns.
- Supports regularization.
- Provides configurable training parameters.

Limitations

- Requires appropriate tuning.
- Can overfit if insufficiently regularized.
- May be more difficult to interpret than simpler algorithms.

XGBoost should be included in the final comparison only if it has been implemented and evaluated successfully.

11.6 Model Comparison Strategy

All candidate models should be evaluated using consistent training data, validation procedures, and performance metrics.

The final model should be selected according to its verified performance and the analytical objective.

No model should be declared the best before its results have been calculated.

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12. MODEL TRAINING AND VALIDATION

Model validation is particularly important because the dataset contains only 303 observations.

A single train-test split may produce performance estimates that vary substantially depending on which observations are selected.

12.1 Training and Testing

The dataset is divided into training and testing subsets.

The training data is used for fitting models, selecting algorithms, and tuning hyperparameters.

The test data is reserved for the final evaluation.

12.2 Stratified Cross-Validation

Stratified five-fold cross-validation can be applied to the training data.

During each iteration, four folds are used for training and one fold is used for validation.

The process is repeated until every fold has served as the validation fold.

The resulting metrics can be averaged to provide an estimate of model performance across different subsets.

Stratification helps preserve the approximate class proportions in each fold.

12.3 Hyperparameter Tuning

Grid Search or Randomized Search can be used to investigate promising model configurations.

Potential parameters include:

- Regularization strength for Logistic Regression.
- Maximum depth for Decision Tree.
- Number of trees and maximum depth for Random Forest.
- Kernel and regularization parameters for SVM.
- Learning rate and tree depth for XGBoost.

Hyperparameter tuning should use the training data and an appropriate cross-validation procedure.

12.4 Final Test Evaluation

After selecting the model using training and validation results, the selected configuration is evaluated on the held-out test set.

The final test results should be reported separately from cross-validation results.

The test set should not be repeatedly used to select models or thresholds.

12.5 Reproducibility

The experiment should record:

- Random seeds.
- Train-test split strategy.
- Cross-validation configuration.
- Preprocessing steps.
- Model hyperparameters.
- Evaluation metrics.
- Python and library versions.

Documenting these details makes the analysis easier to reproduce and review.

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13. MODEL EVALUATION METRICS

Evaluating multiple metrics provides a more complete understanding of classification performance than accuracy alone.

13.1 Accuracy

Accuracy measures the proportion of predictions that are correct.

It considers both correctly predicted positive and correctly predicted negative observations.

However, accuracy does not explain which types of classification errors occur.

13.2 Precision

Precision measures the proportion of predicted positive observations that are actually positive according to the dataset's labels.

Higher precision means fewer false positives relative to the total number of positive predictions.

13.3 Recall

Recall measures the proportion of actual positive observations that the model correctly identifies.

It is particularly useful for examining how many positive cases are missed.

However, increasing recall can also increase the number of false-positive predictions.

13.4 F1-Score

The F1-score combines precision and recall into a single metric.

It is the harmonic mean of these two measures.

It can be useful when both precision and recall are important to the evaluation.

13.5 ROC-AUC

The Area Under the Receiver Operating Characteristic Curve measures how effectively a model ranks positive observations above negative observations across different classification thresholds.

A higher ROC-AUC indicates stronger discrimination within the evaluated data.

13.6 Confusion Matrix

A confusion matrix summarizes four outcomes:

- True positives.
- True negatives.
- False positives.
- False negatives.

It provides a direct way to investigate the types of errors made by a classifier.

13.7 Selecting Evaluation Metrics

The preferred evaluation metric depends on the analytical objective.

Accuracy provides an overall measure of correct predictions.

Precision and recall provide more detailed information about positive predictions and missed positive cases.

F1-score combines precision and recall, while ROC-AUC evaluates ranking performance across thresholds.

For this project, all these metrics should be considered together.

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14. EXPLAINABLE ARTIFICIAL INTELLIGENCE

14.1 Introduction

Explainable Artificial Intelligence, or XAI, refers to methods that help people understand aspects of a machine learning model's behavior.

In this project, explainability is used to investigate how the trained models use the available features to generate predictions.

The goal is to improve analytical transparency rather than establish medical causation.

14.2 Feature Importance

Random Forest provides impurity-based feature importance scores.

These scores indicate how much each feature contributes to reducing impurity across the fitted trees.

A bar chart can be used to display the highest-ranked features.

However, impurity-based importance can be influenced by feature characteristics and correlations between predictors.

Therefore, these scores should not be treated as definitive evidence of medical importance.

14.3 Permutation Feature Importance

Permutation importance measures how model performance changes when the values of one feature are shuffled.

If shuffling a feature substantially reduces performance, the model may rely on that feature.

The method should be applied to appropriate validation or test data, using a clearly documented evaluation metric.

Correlated features can complicate interpretation because multiple predictors may contain overlapping information.

14.4 SHAP Analysis

SHAP provides a framework for estimating how feature values contribute to individual model predictions relative to a defined baseline.

Potential visualizations include:

- SHAP summary plot.
- SHAP feature-importance plot.
- SHAP waterfall plot.

A summary plot can help identify features that have substantial influence on the fitted model across many observations.

A waterfall plot can illustrate how feature contributions move an individual prediction away from the model's baseline output.

The appropriate SHAP explainer depends on the selected algorithm and model configuration.

14.5 Interpretation and Responsibility

Explainability methods can help identify patterns learned by a model.

However, feature importance and SHAP values describe model behavior rather than prove that a feature causes heart disease.

The interpretation should remain specific to the dataset and fitted model.

Implementation note: Feature importance, permutation importance, and SHAP results should be reported as completed findings only after the relevant analyses have been performed.

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15. PROBABILITY AND DECISION THRESHOLD ANALYSIS

15.1 Probability Estimation

Some classification models estimate the probability that an observation belongs to the positive class.

These estimates reflect the patterns learned from the training data.

They are not automatically calibrated medical probabilities.

Calibration must be evaluated separately before probability estimates are interpreted quantitatively.

15.2 Classification Threshold

A classification threshold converts a predicted probability into a class label.

For example, a threshold of 0.50 assigns the positive class when the estimated probability is at least 0.50.

Different thresholds can produce different precision and recall values.

15.3 Threshold Comparison

The analysis can investigate thresholds of 0.30, 0.40, 0.50, 0.60, and 0.70.

For each threshold, the following should be calculated:

- Precision.
- Recall.
- F1-score.
- False positives.
- False negatives.

The calculations must be based on actual model predictions.

15.4 Interpretation

Lower thresholds generally classify more observations as positive. This can increase recall while reducing precision.

Higher thresholds generally classify fewer observations as positive. This can increase precision while reducing recall.

The exact changes depend on the model and dataset.

Threshold selection should be performed using training or validation data rather than repeatedly optimizing against the final test set.

No threshold should be described as medically optimal without appropriate clinical evidence.

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16. RESULTS AND DISCUSSION

16.1 Initial Model Evaluation

The initial analysis compared Logistic Regression and Random Forest using stratified five-fold cross-validation.

The previously reported mean cross-validation results were:

Logistic Regression

- Accuracy: 84.2%.
- Precision: 82.0%.
- Recall: 91.5%.
- F1-score: 86.4%.
- ROC-AUC: 0.890.

Random Forest

- Accuracy: 82.8%.
- Precision: 82.0%.
- Recall: 88.5%.
- F1-score: 85.0%.
- ROC-AUC: 0.909.

These results should be reproduced and verified against the actual notebook or code before final publication.

They represent cross-validation estimates, not final held-out test-set results.

16.2 Preliminary Interpretation

Based on the previously reported results, Logistic Regression achieved higher mean accuracy, recall, and F1-score than Random Forest.

Random Forest achieved a higher mean ROC-AUC.

This illustrates why model evaluation should consider multiple metrics rather than relying on accuracy alone.

The model with the highest accuracy is not necessarily the best model for every analytical objective.

16.3 Extended Model Comparison

The extended analysis should compare Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine.

XGBoost may also be included if implemented.

The comparison should report the verified mean cross-validation results for each model.

The selected model should then be evaluated on the held-out test set.

The discussion should explain the observed differences without assuming that a more complex model must perform better.

16.4 Explainability Results

Once feature importance and SHAP analyses have been performed, the report should identify the features that contribute most to the fitted model's predictions.

The interpretation should explain what the model learned from the supplied dataset while avoiding unsupported claims about causation or medical significance.

16.5 Threshold Analysis Results

The threshold analysis should demonstrate how precision, recall, and classification errors change across the selected thresholds.

These results can help explain the trade-offs involved in converting predicted probabilities into class labels.

Important: Only the initial Logistic Regression and Random Forest metrics have been reported previously. Additional model comparisons, final test results, SHAP findings, and threshold analysis must be calculated before being presented as completed results.

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17. HEALTHCARE ANALYTICS DASHBOARD

17.1 Dashboard Overview

A Power BI dashboard can be developed to present the dataset characteristics, exploratory findings, and model evaluation results.

The dashboard is intended for educational analysis and does not provide medical diagnoses.

It can be organized into three pages.

17.2 Page One: Dataset Overview

This page provides a high-level summary of the dataset.

Suggested components include:

- Total number of observations.
- Number of input features.
- Distribution of the target classes.
- Age distribution.
- Selected feature distributions.
- Data quality summary.

Suggested visualizations include KPI cards, bar charts, histograms, and box plots.

17.3 Page Two: Exploratory Data Analysis

This page focuses on relationships and distributions within the supplied data.

Suggested visualizations include:

- Target distribution by chest pain category.
- Maximum heart rate distribution by target class.
- ST depression distribution.
- Exercise-induced angina by target class.
- Selected feature relationships.
- Correlation heatmap.

Each visualization should include an informative title and a brief interpretation.

The charts should describe observed patterns rather than imply causal relationships.

17.4 Page Three: Model Performance

This page presents the model evaluation results.

Suggested components include:

- Model performance summaries.
- Accuracy, precision, recall, and F1-score.
- ROC-AUC comparison.
- Confusion matrices.
- ROC curves.
- Feature importance visualization.
- Threshold comparison.

Cross-validation results and final test-set results must be clearly distinguished.

The dashboard should not display invented metrics or unverified model outputs.

17.5 Dashboard Design Principles

The dashboard should use consistent formatting, readable labels, and appropriate chart types.

The visualizations should distinguish descriptive dataset statistics from model predictions.

If an example prediction interface is included, it must clearly state that the output is educational and not a medical diagnosis.

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18. KEY FINDINGS

The initial dataset inspection established the following observations:

- The supplied dataset contains 303 observations and 13 input features.
- The target variable contains 138 observations in class 0 and 165 observations in class 1.
- No missing values were detected in the supplied CSV.
- Logistic Regression and Random Forest were evaluated as initial classification models.
- The previously reported cross-validation results show different performance patterns across evaluation metrics.

The following findings should be added after the extended experiments are completed:

- The best-performing model under the selected evaluation criterion.
- The variation in performance across cross-validation folds.
- The final held-out test-set performance.
- The most influential features according to the selected explainability methods.
- The effects of different classification thresholds.
- The limitations affecting generalizability.

All final conclusions must be supported by calculated results.

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19. LIMITATIONS

19.1 Small Dataset

The dataset contains only 303 observations.

This limits the precision of performance estimates and the ability to represent diverse populations.

19.2 Dataset Representativeness

The dataset may not represent contemporary healthcare settings or all demographic groups.

The model's performance may differ when applied to another population.

19.3 Target Definition

The exact meaning and encoding of the target variable must be verified using the dataset documentation.

Incorrect assumptions about the labels could lead to misleading interpretations.

19.4 Model Generalization

A model may perform well on the supplied dataset but perform differently on independent data.

External validation would be required to assess its performance beyond the original sample.

19.5 Potential Bias

The project does not provide a comprehensive evaluation of subgroup performance or fairness.

Further investigation would be needed to understand potential differences across demographic groups.

19.6 Probability Calibration

Predicted probabilities are not necessarily calibrated.

Calibration analysis would be required before treating them as reliable estimates of outcome likelihood.

19.7 Explainability Limitations

Feature importance and SHAP explanations describe model behavior.

They do not establish causal relationships or prove the clinical significance of individual features.

19.8 Clinical Limitations

The models have not been clinically validated.

They must not be used to diagnose heart disease, recommend treatment, or reassure individuals about their health.

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20. FUTURE SCOPE

The project can be extended through several improvements.

20.1 Advanced Model Comparison

Additional algorithms and carefully selected hyperparameter configurations can be evaluated under consistent validation conditions.

20.2 Hyperparameter Optimization

Grid Search or Randomized Search can be used to identify promising model configurations.

The optimization process should use training data and cross-validation.

20.3 External Validation

An independent dataset could be used to investigate whether the selected model generalizes beyond the original sample.

20.4 Probability Calibration

Calibration methods and reliability diagrams could be used to assess the relationship between predicted probabilities and observed outcomes.

20.5 Advanced Explainability

SHAP and permutation importance can be supplemented with additional interpretation techniques.

20.6 Robustness Analysis

The analysis could investigate how performance changes under different data splits and reasonable preprocessing choices.

20.7 Interactive Reporting

The Power BI dashboard could be expanded to provide additional exploratory visualizations and model comparison summaries.

20.8 Reproducible Deployment

The project could be packaged with a requirements file, reproducible notebooks, and clear execution instructions.

A future deployment should remain educational unless appropriate clinical validation, governance, and regulatory requirements have been satisfied.

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21. CONCLUSION

This project presents an end-to-end approach to analyzing a heart disease dataset using machine learning and explainable analytics.

The workflow includes dataset inspection, exploratory data analysis, preprocessing, model development, cross-validation, performance evaluation, and interpretation.

Logistic Regression and Random Forest provide the initial baseline comparison. Additional classification algorithms and explainability methods can extend the analysis and provide a broader understanding of model behavior.

The project demonstrates that evaluating a classification model requires more than measuring accuracy.

Precision, recall, F1-score, ROC-AUC, confusion matrices, and validation variability provide complementary information about predictive performance.

Explainability techniques can help investigate how the fitted models use the available features, while appropriate validation reduces the risk of overestimating performance.

The project also emphasizes the importance of communicating healthcare-related analytical findings responsibly.

Overall, the case study provides a foundation for a reproducible data science portfolio project combining machine learning, model evaluation, explainability, and visualization.

The findings are limited to the supplied dataset and should not be interpreted as evidence of clinical effectiveness.

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22. REFERENCES

The following resources provide documentation for the technologies and methods used in the project.

Scikit-learn Documentation

https://scikit-learn.org/stable/

Pandas Documentation

https://pandas.pydata.org/docs/

NumPy Documentation

https://numpy.org/doc/

Matplotlib Documentation

https://matplotlib.org/stable/

Seaborn Documentation

https://seaborn.pydata.org/

SHAP Documentation

https://shap.readthedocs.io/

XGBoost Documentation

https://xgboost.readthedocs.io/

Microsoft Power BI Documentation

https://learn.microsoft.com/power-bi/

The specific source and version of the `heart.csv` dataset should also be cited in the final report. Its provenance and target definition should be verified before making claims about how the observations were collected or what the target represents.

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23. PROJECT DISCLAIMER

This project is intended for educational and portfolio purposes only.

The dataset is small, and the models have not been clinically validated.

The model outputs must not be used as a substitute for professional medical advice, diagnosis, or treatment.

Feature importance and explainability results describe model behavior and do not establish medical causation.

The project does not demonstrate that the models can reliably diagnose heart disease or predict an individual's clinical risk.

Any real-world healthcare application would require appropriate data governance, independent validation, clinical evaluation, and regulatory review where applicable.

PROJECT SUMMARY

Project Title: Heart Disease Risk Prediction and Explainable Healthcare Analytics

Domain: Healthcare Analytics and Data Science

Dataset: heart.csv

Programming Language: Python

Core Libraries: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn

Optional Explainability Tool: SHAP

Machine Learning Algorithms: Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, and optionally XGBoost.

Key Techniques:

- Exploratory Data Analysis.
- Data Preprocessing.
- Binary Classification.
- Stratified Cross-Validation.
- Hyperparameter Tuning.
- Classification Metrics.
- Feature Importance.
- Explainable AI.
- Probability Threshold Analysis.
- Data Visualization.

Primary Deliverable: A documented machine learning analysis supported by reproducible code, verified evaluation results, explainability visualizations, and an optional Power BI dashboard.

Final Objective: Demonstrate a systematic approach to analyzing healthcare-related data, evaluating predictive models, interpreting model behavior, and communicating findings responsibly.